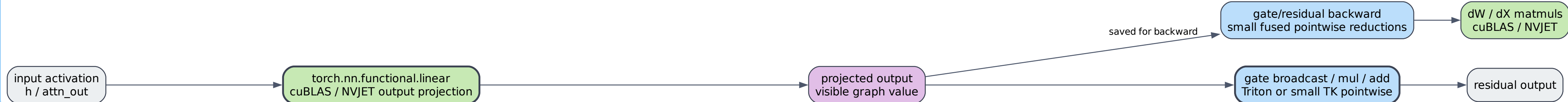


Output Projection Fusion Boundary: TK Custom vs torch.compile-visible Epilogue

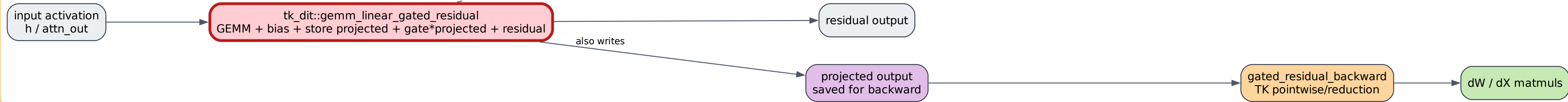
Epilogue-visible path: compile_fused_output_proj_epilogue



Measured Finding

Custom projected gated GEMM appears as hot TK kernel:
~1.514 ms over 72 calls in short profiler run.
ptxas reports 24-28 B spills for gated_linear_template.
Epilogue-visible path removes that kernel and lets cuBLAS/NVJET own GEMM.

Current TK fused path: compile_fused_output_proj



Legend

Register spill observed

Saved tensor for backward

TK custom fused kernel

torch.compile fused Triton pointwise

GEMM / matmul